

Machine learning for mangrove species distribution using Sentinel 2 satellite image in Segara Anakan, Cilacap Region, Indonesia

Sigit Febrianto^{a,b,*}, Arif Rahman^a, Oktavianto Eko Jati^a, Anindya Wirasatriya^c,
Max Rudolf Muskananfolo^a, Nurul Latifah^a

^a Department of Aquatic Resources, Faculty of Fisheries and Marine Science, Universitas Diponegoro, Kampus FPIK UNDIP, Jl. Prof. Sudharto, SH, Tembalang, Semarang 50275, Indonesia

^b Sigit Blue Carbon Laboratory, Universitas Diponegoro, Kampus FPIK UNDIP, Jl. Prof. Sudharto, SH, Tembalang, Semarang 50275, Indonesia

^c Department of Oceanography, Faculty of Fisheries and Marine Science, Universitas Diponegoro, Kampus FPIK UNDIP, Jl. Prof. Sudharto, SH, Tembalang, Semarang 50275, Indonesia

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ABSTRACT

Indonesia is one of the countries with the most extensive mangrove forests in the world. They are spread across the Papua, Sulawesi, Kalimantan, Sumatra and Java islands. Segara Anakan is the largest natural mangrove on Java, draining into the Indian Ocean. This study uses Sentinel 2 satellite imagery from 2022. The advantage of using this satellite data is that it is freely available and has a spatial resolution of 10 m. It can be used to map mangrove species. Mangrove species mapping combines data and machine learning random forest and support vector machine, while mangrove vegetation density analysis uses the Normal Difference Vegetation Index (NDVI). Based on the spatial classification, 13 mangrove species were identified as dominant in the study area. These include *Xylocarpus moluccensis*, *Xylocarpus granatum*, *Sonneratia caseolaris*, *Sonneratia alba*, *Rhizophora stylosa*, *Rhizophora mucronata*, *Rhizophora apiculata*, *Nypa fruticans*, *Excoecaria agallocha*, *Ceriops decandra*, *Bru-guiera gymnorhiza*, *Aveennia marina* and *Aegiceras corniculatum*. The accuracy test results using a confusion matrix with a window size of 15×15 showed an overall accuracy (OA) of 91.28 %. On the western side of Segara Anakan, *Xylocarpus moluccensis*, *Xylocarpus granatum*, *Rhizophora mucronata* and *Aveennia marina* were found. The species that are widely distributed in this area are *Nypa fruticans*, covering 4938.25 ha, and *Rhizophora apiculata*, covering 1621.03 ha with high density, while the least distributed is the species of *Excoecaria agallocha*, covering 3.33 ha. The results of mangrove species distribution research using machine learning can be used to support mangrove rehabilitation conservation according to natural mangrove type. The novelty of this research can be useful for monitoring mangrove species and mangrove conservation, especially in the Segara Anakan mangrove forest area.

1. Introduction

Mangrove forests are coastal ecosystems that can adapt to a wide range of salinities and are highly productive. They support fauna, provide nutrients to coastal waters, and support local economies (Alongi, 2018; Sidik et al., 2023; Giri et al., 2011). Mangroves grow and spread along coastlines, estuaries, bays and brackish water (Alongi and Mukhopadhyay, 2015; Rhyma et al., 2020; Ahmed et al., 2021). Mangrove forests have a physical role in preventing coastal erosion and wave action, as well as a biological role as nesting and nursery grounds

for biota to support biodiversity. Still, they are under tremendous pressure, with the world's mangrove area regularly declining due to erosion, human activities, logging and shrimp farming (Hutchison et al., 2014; Boone Kauffman et al., 2017). Unlike terrestrial vegetation, mangroves have a unique rooting structure, with prop roots and pneumatophores in the soil hanging in the air from tree branches. This allows this ecosystem to trap large amounts of sediment. With this root system, mangroves can grow forward and backwards depending on the suitability of the primary substrate that supports mangrove growth and development (UNESCO, 2017; Kazemi et al., 2021).

* Corresponding author at: Department of Aquatic Resources, Faculty of Fisheries and Marine Science, Universitas Diponegoro, Kampus FPIK UNDIP, Jl. Prof. Sudharto, SH, Tembalang, Semarang 50275, Indonesia.

E-mail addresses: sigitfebrianto40@lecturer.undip.ac.id, febriantosigit40@gmail.com (S. Febrianto).

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Indonesia has the largest mangrove area in the world, with an area of 3112,989 ha, equivalent to 22.6 % of the global mangrove area. Mangroves are spread in several regions, including Irian Jaya, Sulawesi, Sumatra, Borneo, and Java. One of the mangroves in Java that is widespread and still in natural conditions is in Segara Anakan Cilacap (Febrianto et al., 2022; Giri et al., 2011; Latifah et al., 2018; Murdiyarso et al., 2015). Segara Anakan is located on the southern coast of Central Java (108°50'3.26" E, - 07°40'23.45" S) and faces the Indian Ocean and Nusakambangan Island (Fig. 1). The Segara Anakan mangrove forest is the largest and most extensive natural mangrove forest in the Java Island. With very high biodiversity, the Segara Anakan mangrove plays an important ecological role as a migration route for silver eels *Anguilla bicolor*, a catadromous organism (Arai and Chino, 2022). Four major rivers, Cintandui, Cibeureum, Jeruk Legi and Tritih, drain into the Segara Anakan mangrove forest. The total area of mangroves in Segara Anakan decreased from 13,557 ha in 1997–8036.9 ha in 2012 due to land conversion. Segara Anakan mangroves are threatened by anthropogenic factors and sediment erosion in upland areas originating from the Citanduy watershed, with an estimated amount of sediment deposited at 1000,000 m³ / year. This will make the Segara Anakan lagoon area increasingly shrinking, and high sediment input can cover the mangrove root system, especially pneumatophores if many are submerged in elevated amounts of sediment, it can cause the death of mangroves and threaten the diversity of flora and fauna that live in association with mangroves (Cahyo et al., 2024; Hakiki et al., 2021; Kaliraj et al., 2017). Another anthropogenic factor in Segara Anakan is increased urbanisation and immigration since 1987. This has resulted in many areas being converted into agricultural land, rice fields, shrimp farms, fish farms, chicken farms, and fishermen's settlements of Kampung Laut. In Segara Anakan, there are several types of mangroves, with the most significant area such as *Nypa fruticans*, *Rhizophora apiculata*, *Agiceras corniculatum* and *Aveenia marina*, and the substrate beneath these mangroves is a habitat for eels—*Anguilla bicolor* which is a protected species under the IUCN Red List category.

Considering the importance of mapping mangrove ecosystem types in Segara Anakan can be used as efforts should be made to restore the degraded mangrove forest. Damage to mangrove forests due to land clearing and sedimentation initiated mangrove rehabilitation efforts

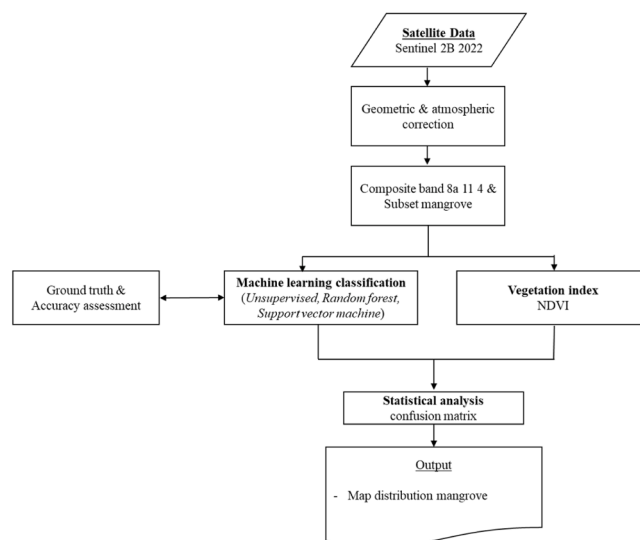


Fig. 2. Flowchart of the research.

that have been carried out where the types of mangroves planted are dominated by *Rhizophora apiculata*, *Rhizophora mucronate* and *Bruguiera gymnorhiza*. In contrast, for other mangrove species, no nursery efforts have been made, while areas that have not been restored are dominated by shrubs, namely *Acanthus ilicifolius* and *Derris trifoliata* (Ardli et al., 2015; Isdianto and Luthfi, 2021). Mangrove restoration in the first three months faces many challenges from crabs and disease. Rehabilitation using single-species mangroves, which are not the species of origin, is susceptible to disease attacks and can lead to mass mangrove die-offs such as those that have occurred in Sulawesi, Borneo and Vietnam (Jiang et al., 2022; Maryam et al., 2018).

Several studies related to mangrove ecosystems in Segara Anakan have been carried out about changes in the mangrove area, comparison of the mangrove index and Normalized Difference Vegetation Index (NDVI) and the impact of mangrove deforestation. The weakness of some previous studies is that they can only distinguish mangroves from

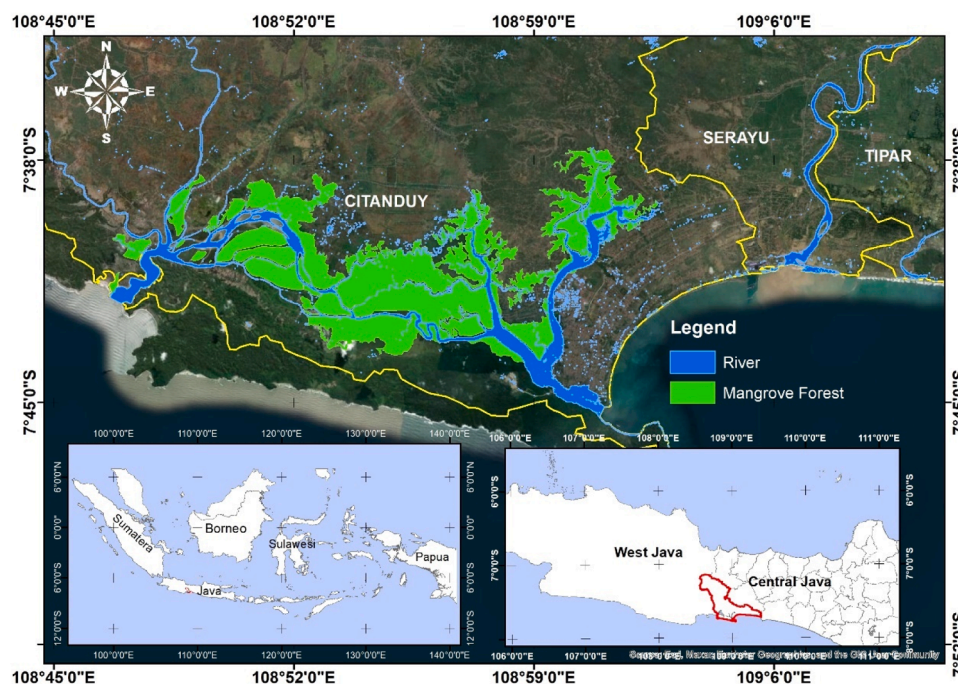


Fig. 1. Study site map, located in Segara Anakan, Cilacap Regency, Central Java Province, Indonesia.

non-mangroves and their density spatially, but there is no research related to the spatial distribution of mangrove species in Segara Anakan using machine learning (Isdianto and Luthfi, 2021; Nordhaus et al., 2019; Winarso et al., 2023).

On the other hand, the spatial analysis of mangrove species using the machine learning method has been conducted in Rembau River, Malaysia, using the maximum likelihood classifier method that can separate eight classes of mangrove species with an overall accuracy of 84.81 %. The study results serve to monitor and conserve mangroves as firefly habitat. In the present study, we first apply machine learning to determine the distribution of mangrove species in Segara Anakan and determine the types of species that can survive under the pressure of sedimentation in the Citanduy watershed.

The results of the study will be helpful for government agencies, NGOs and monitoring groups to carry out mangrove rehabilitation based on the distribution of natural species.

2. Materials and methods

2.1. Satellite data

The Sentinel 2 satellite, launched by the European Space Agency (ESA), is a consortium of satellites from European countries consisting of two Sentinel 2 A and two Sentinel 2B satellites with 13 bands. The Sentinel 2 satellite has the advantage of being freely accessible and has a spatial resolution of 10 m and a temporal resolution of 10 days, which is better than other satellites, such as Landsat 8, with three different resolutions of 10 m, 20 m and 60 m and a time resolution of 10 days (Drusch et al., 2012; European Space Agency, 2012; Prakash Hati et al., 2021). Multi-Spectral Instrument (MSI) used in 2022 with temporal resolution owned by this satellite is ten days with the highest spatial resolution of 10 m contained in B2, B3, B4, and B8 (Roy et al., 2016). The satellite data were processed using Qgis 3.16 and ArcGIS 10.8 software.

2.2. Ground truth data

Field data was collected using a Global Positioning System (GPS) Garmin 78 s. One hundred thirty-six points were recorded, consisting of data on mangrove species, water bodies and built-up areas, which will be used for classification references. Field data on mangrove species and diameter at breast height were collected from 10 x 10 m transect plots from 18 randomly distributed sites from upstream to downstream. In total, 13 types of mangrove species have been identified, including *Xylocarpus moluccensis*, *Xylocarpus granatum*, *Sonneratia caseolaris*, *Sonneratia alba*, *Rhizophora stylosa*, *Rhizophora mucronata*, *Rhizophora apiculata*, *Nypa fruticosa*, *Excoecaria agallocha*, *Ceriops decandra*, *Bruguiera gymnorhiza*, *Aveicennia marina*, *Aegiceras corniculatum* and there are two additional classes of water and built-up area.

2.3. Data analysis

2.3.1. Image processing

The Sentinel 2 satellite image used is a Level 1 C product in jp2 data format. The image has been subjected to radiometric, atmospheric and geometric corrections using TOA. The image was projected using the World Geodetic Survey and Universal Transverse Mercator (UTM) 49S datum. Sentinel 2B images were subset according to the research site. The composite bands 8 A, 11, 4 and 5 have been used to identify mangroves and non-mangroves (Baloloy et al., 2020). Optimum Index Factor (OIF) image enhancement for improved image quality (Chavez and Bauer, 1982). The OIF can be described using the following formula:

$$OIF = \frac{\sum (i=1) 3SD_i}{\sum (j=1) 3ABS|CC_j|} \quad (1)$$

Where SDi = standard deviation and ABS CCj = absolute correlation coefficient.

2.3.2. Normal Difference Vegetation Index (NDVI)

NDVI is an indicator of the greenness of plants that is used to separate the information on vegetation and non-vegetation based on the red band and the NIR band, which is written in the equation below (Cissell et al., 2018):

$$NDVI = (NIR - RED) / (NIR + RED) \quad (2)$$

Where NIR is a strong spectral reflectance that absorbs the green colour of plants in healthy and dense conditions by 50 %, while less healthy plants are absorbed by 40 %, and Red is a spectral reflectance at visible red wavelengths with a range of values from -1 to 1 (Ghebregabher et al., 2020; Das, 2021). NDVI values > 0.25 indicate the presence of vegetation. Values -1 < 0 indicate the absence of vegetation, which can be water bodies, settlements and open land (Rhyma et al., 2020). Using the NDVI values, we can separate the mangroves' density by dividing the values' range.

2.3.3. Image classification

The research used three image classification types: unsupervised, random forest and support vector machine. There is a fundamental difference between the three classification techniques. The number of training data samples used was as many as 169, with each mangrove species having as many as 11–12 polygons, and the software used to process them was ArcGIS 10.8. Unsupervised algorithms classify objects without knowing the type of vegetation or land cover based on the digital number of each piece of information contained in the image pixel (Cissell et al., 2018). There are 5 land use classes based on an object's digital number characteristics (Fig. 3). The RF algorithm is an early-stage machine learning method that attempts to apply a non-parametric classification approach that requires reference data or training samples to create a new classification on satellite images by identifying vegetative-type objects based on field data (Fig. 4) as a reference for data validation. (Ilham and Marzuki, 2018; Jachowski et al., 2013). Support Vector Machine algorithm (SVM) is a development of supervised learning that can be applied to image classification with linear or non-linear regression approach (Fig. 5). This algorithm can find the best hyperplane in N-dimensional space so that it can separate between object classes better because it can read the class in two dimensions by producing line while and three dimensions produce plane similar (Xu et al., 2023; Dang et al., 2019). The regression formula based on the sample training can be written in the following equation $D = \{(x_1, y_1), (x_2, y_2), \dots, (x_m, y_m)\}$, We hope to learn a regression model $f(x) = wTx + b$ such that $f(x)$ is as close to y as possible. The objective function is given by:

$$\min \frac{1}{2} \|\omega\|^2 + C \sum_{i=1}^m l(f(x_i) - y_i) \quad (3)$$

The SVM is divided into linear and nonlinear kernels, where ω and b are model parameters.

2.3.4. Accuracy assessment

Estimating accuracy using confusion matrix analysis is done to determine the model's accuracy based on the category between mangrove species. The diagonal rows indicate pixels of mangrove species that are correctly identified based on the reference data, while pixels of mangrove species outside the diagonal indicate misclassification of species. The overall accuracy (OA), producer's accuracy (PA), user's accuracy (UA) and kappa coefficient (K) are some of the outputs used to determine the accuracy of the test (Hoque et al., 2021; Mishra et al., 2021). The total number of observation points collected in the field for model building is 160 points samples of mangrove species types and 9 points samples of water bodies and built-up areas. A comparison of

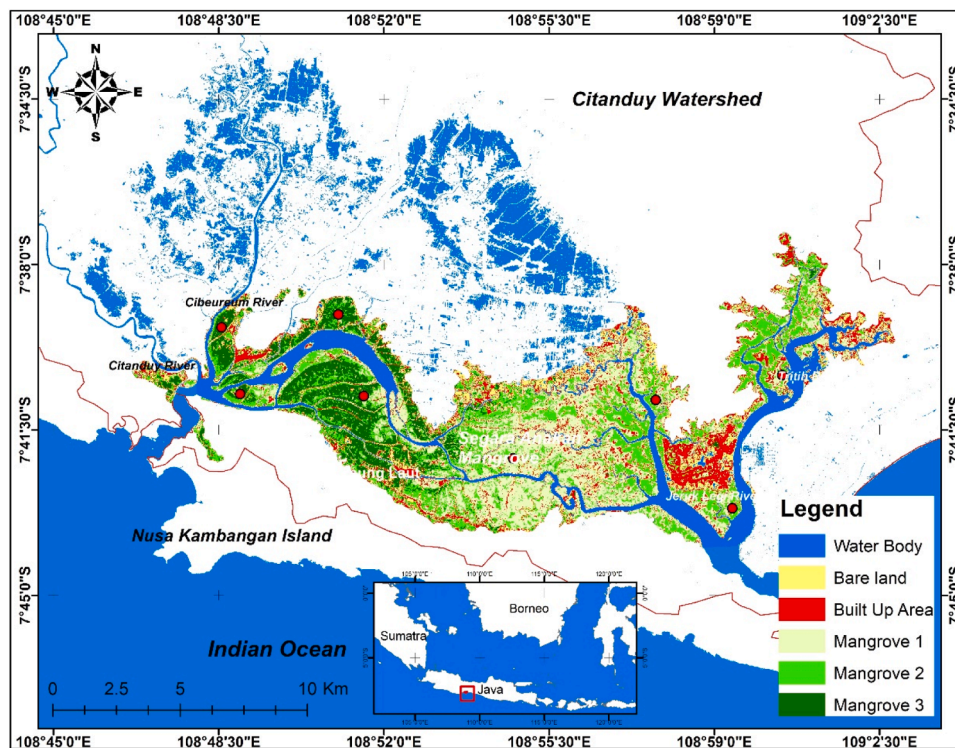


Fig. 3. Unsupervised classification of mangrove at Segara Anakan Cilacap in 2022.

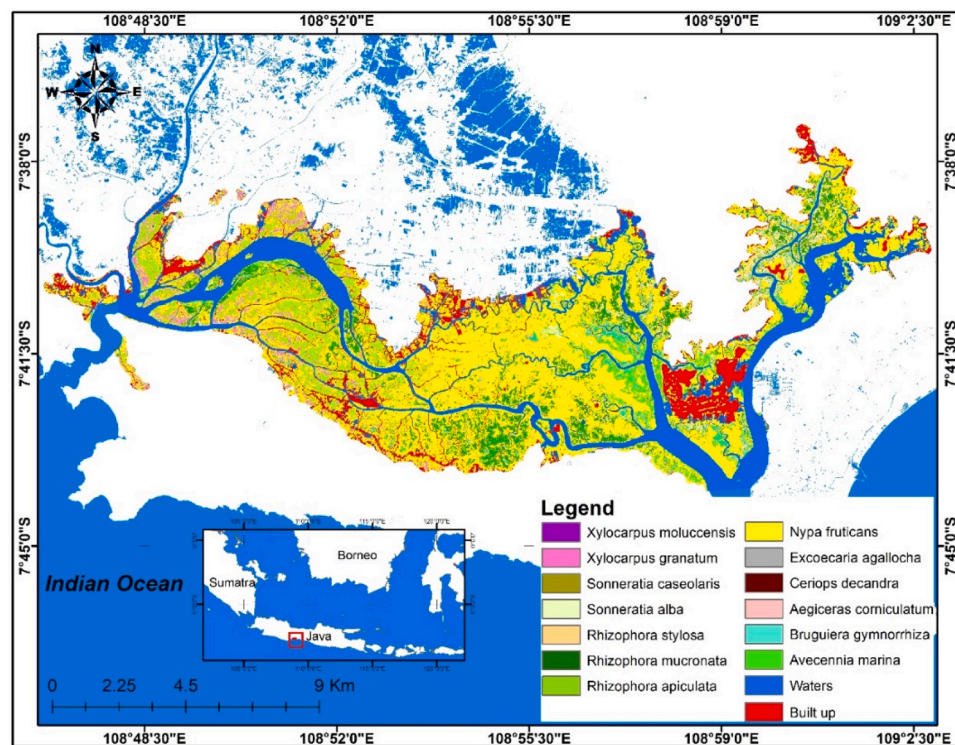


Fig. 4. Random forest classification of mangrove at Segara Anakan Cilacap in 2022.

the ability and performance of the results of unsupervised classification, random forest (RF), support vector machine (SVM), and NDVI in separating mangrove vegetation and mangrove species distribution in Segara Anakan based on field data is presented in Fig. 4 and Fig. 5.

3. Results and discussion

3.1. Mangrove distribution from machine learning classification

Classification of mangrove species distribution in Segara Anakan

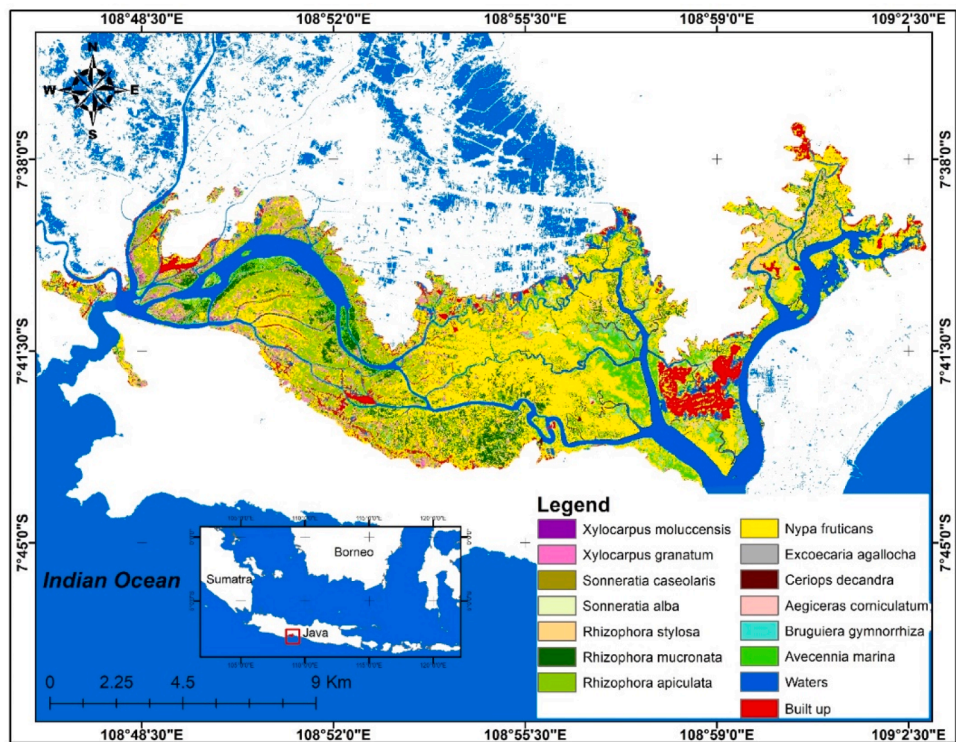


Fig. 5. Support vector machine classification of mangrove at Segara Anakan Cilacap in 2022.

based on Sentinel-2 satellite imagery using unsupervised machine learning method analysis resulted in 3 mangrove categories: mangrove 1, mangrove 2, mangrove 3 (Fig. 3) showing mangrove distribution and land use with 6×6 window size. Thus, species distribution cannot be obtained by unsupervised machine learning method. This is due to the difference in the underlying principles, where unsupervised is used to find patterns based on the digital number of images without needing training data, whereas other classification analyses use training data to predict the classification that will occur based on the amount and distribution of training data. On the other hand, RF classification analysis shows better results that can identify 13 dominant mangrove species with a 15×15 window size (Fig. 4). They are randomly distributed with different mangrove species areas. The largest mangrove area is *Nypa fruticans*, spread near the river and on the newly formed sedimentation deposits. The smallest area covered is *Rhizophora* species. Similarly, SVM classification analysis revealed 13 dominant mangrove species with a window size of 15×15 . They are *Xylocarpus moluccensis*, *Xylocarpus granatum*, *Sonneratia caseolaris*, *Sonneratia alba*, *Rhizophora mucronata*, *Rhizophora apiculata*, *Nypa fruticans*, *Excoecaria agallocha*, *Ceriops decandra*, *Aegiceras corniculatum*, *Bruguiera gymnorhiza*, *Aveicennia marina*. This mangrove distribution is in agreement with Ramírez-vargas et al., (2023) and Sejati et al., (2020). The difference between RF and SVM analysis is the spatial distribution pattern and extent of each mangrove species in the Segara Anakan area, especially the distribution of *Rhizophora* mangrove species. The spatial distribution of mangrove species is also determined by the amount of field data and the random distribution of data to help identify existing mangrove species based on pattern and distance (Abdel-Hamid et al., 2018; Gilani et al., 2021; Gyamfi-Ampadu et al., 2020).

To ensure the best result of the RF and SVM methods, we performed a confusion matrix statistical analysis on 169 sample polygons. We examined the accuracy of the 15 classes, which consisted of 13 types of mangrove species and 2 other land uses (water and built-up areas). The main objective of measuring the accuracy of the confusion matrix was to produce a classification with the highest level of accuracy. The RF method has the advantage of performing vegetation classification by

combining several simple prediction models that operate independently using a decision tree approach based on randomly sampled data. Still, it also has the disadvantage of data interpretation that is limited to the features being classified, the distribution data and the distribution of field parameters, and data with a very complex class matrix (Aji et al., 2023). The resulting accuracy of the RF method is an overall accuracy of 83.25 %, with a kappa statistic of 79.8 %. This accuracy exceeds the minimum classification requirement of 75 %.

The overall accuracy value for the SVM method is 91.28 %, with a kappa statistic of 90.15 % (Table 1). UA shows how high and low the accuracy value of the classified species is based on the results of the analysis of the highest value contained in the species (Ilham and Marzuki, 2018; Pham and Brabyn, 2017). *Xylocarpus moluccensis*, *Xylocarpus granatum*, *Sonneratia caseolaris*, *Sonneratia alba*, *Rhizophora apiculata*, *Bruguiera gymnorhiza*, *Aegiceras corniculatum* with a UA of 100 %. The lowest classification accuracy was found for *Excoecaria agallocha* and *Nypa fruticans*, with UA 66.67 % and 79.31 %, respectively. The results of the accuracy evaluation (Table 2) showed high classification accuracy of mangrove species with a 15×15 classification matrix measuring FoM, O, C and A. The following mangrove species were classified with the highest FoM (over 80 %). *Xylocarpus moluccensis*, *Xylocarpus granatum*, *Sonneratia caseolaris*, *Sonneratia alba*, *Rhizophora stylosa*, *Rhizophora mucronata*, *Ceriops decandra*, *Aveicennia marina* and the lowest FoM over 67 %, while *Rhizophora apiculata* had the lowest. Omission also shows a perpendicular relationship with FoM. The lower the omission value, the higher the FoM, indicating that the species has high accuracy.

Table 1
Sentinel 2 imagery.

Image	Sentinel 2
Scene	S2BMSIL1C
Date	2022/06/30
Time	09.15 am
UTM	49S
Coordinate System	WGS 1984

Table 2

Accuracy of image classification with window size of 15×15 ; UA, user accuracy; PA, producer accuracy; OA, overall accuracy; K, kappa. (Xm = *Xylocarpus moluccensis*, Xg = *Xylocarpus granatum*, Sc = *Sonneratia caseolaris*, Sa = *Sonneratia alba*, Rs = *Rhizophora stylosa*, Rm = *Rhizophora mucronata*, Ra = *Rhizophora apiculata*, Nf = *Nypa fruticosa*, Ea = *Excoecaria agallocha*, Cd = *Ceriops decandra*, Bg = *Bruguiera gymnorhiza*, Am = *Avicennia marina*, Ac = *Aegiceras corniculatum*).

Class name	Xm	Xg	Sc	Sa	Rs	Rm	Ra	Nf	Ea	Cd	Built Up	Bg	Am	Waters	Ac	Total	UA (%)
Xm	4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4	100
Xg	0	10	0	0	0	0	0	0	0	0	0	0	0	0	0	10	100
Sc	0	0	4	0	0	0	0	0	0	0	0	0	0	0	0	4	100
Sa	0	0	0	4	0	0	0	0	0	0	0	0	0	0	0	4	100
Rs	0	0	0	1	10	0	0	0	0	0	0	0	0	0	0	11	90.91
Rm	0	0	0	0	0	21	1	0	0	0	0	0	0	0	0	22	95.45
Ra	0	0	0	0	0	0	5	0	0	0	0	0	0	0	0	5	100
Nf	0	0	0	0	1	0	3	23	0	1	0	0	0	0	1	29	79.31
Ea	0	0	0	0	0	0	0	0	2	0	0	1	0	0	0	3	66.67
Cd	0	0	0	0	0	0	0	0	0	8	0	0	0	0	0	8	100
Built Up	0	0	0	0	0	0	0	1	0	0	10	0	0	2	0	13	76.92
Bg	0	0	0	0	0	0	0	0	0	0	0	3	0	0	0	3	100
Am	0	0	0	0	0	0	0	0	0	0	0	0	4	1	0	5	80.00
Waters	0	0	0	0	0	0	0	0	0	0	0	0	0	27	0	27	100
Ac	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	100
Total	4	10	4	5	11	21	9	24	2	9	10	4	4	30	2	149	
PA (%)	100	100	100	80	90.91	100	55.56	95.83	100	88.89	100	75	100	90	50	136	
OA: 91.28 %																	
K: 90.15 %																	

Table 3

Image classification accuracy with 15×15 window size; FoM, Figure of merit; O, omission; C, commission; A, overall agreement. (Xm = *Xylocarpus moluccensis*, Xg = *Xylocarpus granatum*, Sc = *Sonneratia caseolaris*, Sa = *Sonneratia alba*, Rs = *Rhizophora stylosa*, Rm = *Rhizophora mucronata*, Ra = *Rhizophora apiculata*, Nf = *Nypa fruticosa*, Ea = *Excoecaria agallocha*, Cd = *Ceriops decandra*, Bg = *Bruguiera gymnorhiza*, Am = *Avicennia marina*, Ac = *Aegiceras corniculatum*).

Class name	FoM (%)	O (%)	C (%)
Xm	1	0.00	0.00
Xg	1	0.00	0.00
Sc	1	0.00	0.00
Sa	0.82	20.00	0.00
Rs	0.83	9.09	9.09
Rm	0.95	0.00	4.55
Ra	0.67	44.44	34.20
Nf	0.78	4.17	20.69
Ea	0.73	0.00	33.33
Cd	0.89	11.11	0.00
Built Up	0.79	0.00	23.08
Bg	0.78	25.00	0.00
Am	0.82	0.00	20.00
Waters	0.90	10.00	0.00
Ac	0.64	50.00	0.00
A (Overall agreement)	91.28		

Mangrove species have the highest commission error value, *Rhizophora apiculata* dan *Excoecaria agallocha*, with a value of over 30 % error caused by the low value of FoM, thus increasing the high commission error. Based on the comparison of accuracy results. In conclusion, the SVM method has better accuracy than the RF method. Thus, we use the results of the SVM method for the next analysis.

Based on the results show that the RF and SVM classifications separate the distribution of mangrove species up to 13 classes and non-mangrove species up to 2 classes but produce different areas and different accuracy values where RF classification has the lowest accuracy of 75 % for *Nypa fruticosa* with an area of 5640.65 Ha and *Avicennia marina* has the highest accuracy of 95 % with an area of 418.19 Ha. (Fig. 5) Support vector machine (SVM) classification has the lowest accuracy of 66 % for *Nypa fruticosa* with an area of 4938.24 ha and mangrove species. *Xylocarpus moluccensis*, *Xylocarpus granatum*, *Sonneratia caseolaris*, *Sonneratia alba*, *Rhizophora stylosa*, *Rhizophora mucronata*, *Rhizophora apiculata*, *Bruguiera gymnorhiza* with an accuracy of 90.91–100 %. *Rhizophora apiculata* became the mangrove species with the highest accuracy of 100 % and the highest area of 1621.03 ha.

The resulting mangrove distribution pattern has similarities with the distribution of mangrove species observation coordinates in the field, so the high accuracy value is directly proportional to the actual conditions.

Based on the distribution of species there are specific characteristics of mangrove zoning in Segara Anakan, *Xylocarpus moluccensis* and *Xylocarpus granatum* are scattered in locations close to the mainland and away from the river indicated by purple colour where the most scattered near the island of Nusa Kambangan and in the middle of the island arises near Plawangan Barat of the river mouth. *Nypa fruticosa*, *Aegiceras corniculatum* dominate the distribution of mangroves in the area near the River shown in yellow and pink colours. This is in accordance with the research (Nordhaus et al., 2019) which states that there is an effect of mangrove deforestation based on the area distribution of mangrove growth (Hu et al., 2018; Nordhaus et al., 2019; Purwanto and Asri-ningrum, 2019).

Research (Lombard et al., 2023) on mangrove zonation in West Africa shows three main zonations near estuary, centre, and land with mud bottom substrate. *Rhizophora racemosa* is a high-density mangrove distributed near the mouth of the river; low to medium-density mangroves are dominated by other mangrove species. *Rhizophora mangle*, low to open mangrove density dominated by mangrove species *Rhizophora mangle* and *Avicennia germinans*.

NDVI analyses mangrove cover in Segara Anakan, as shown in (Fig. 6). Based on NDVI analysis, positive NDVI values > 0.3 indicate areas of vegetation. In contrast, negative values -1 – -0.2 indicate areas of non-vegetation, which in this study include embossed land, open areas, settlements, ponds and water bodies. NDVI classification based on the range of values refers to research (Idris et al., 2022; Latifah et al., 2018) which divides NDVI values into 5 classifications based on the object of observation and vegetation density. The pixels of the very high-density class have a value of > 0.6 , indicating a high cover of mangrove vegetation found in the area of Cibereum River and the northern side of Kampung Laut, 0.4 – 0.6 as the high-density class is the highest density class (Etemadi et al., 2021; Myeong et al., 2006) 0.2 – 0.4 shows that the density class transitions from a low to a high-density class composed of vegetation around settlements, sedimentation, and open land. 0 – 0.2 shows a low-density class dominated by land use in the form of settlements and open land. -0.21 – 0 shows the water class because there is no vegetation cover in this range of values. This can be seen in the river channel and new land on the river bank; yellow indicates the accumulation of sediments. (Ghebregabher et al., 2020; Chen et al., 2018). The most dominant scattered NDVI values are indicated by a light

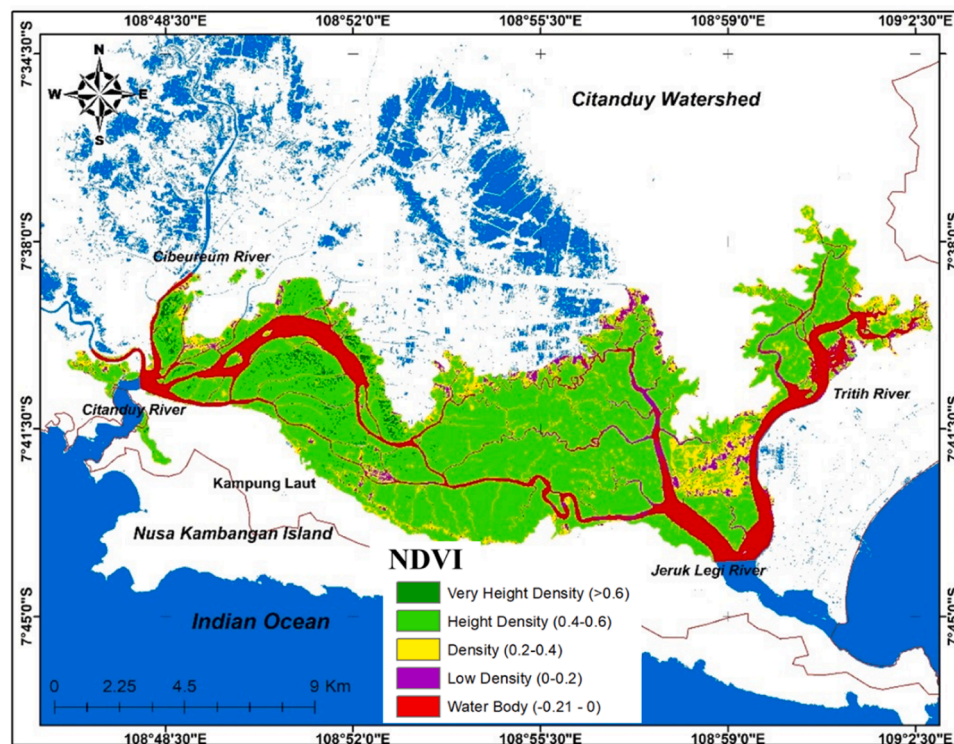


Fig. 6. NDVI Map of mangrove Segara Anakan Cilacap in 2022.

green colour (Fig. 6). Based on the classification results, vegetation index and composite band analysis can detect the presence of mangroves and non-mangroves in detail and the chlorophyll-a content of plants (Kamal et al., 2021).

3.2. NDVI and density of mangrove species

The function of NDVI analysis to distinguish vegetation and non-vegetation cover of Segara Anakan mangroves is overlaid with mangrove species density (Fig. 7a). Research results (Hanggara et al., 2021; Murdiyarso et al., 2021) reported that mangroves planted on the

north coast of Java and Segara Anakan are dominated by *Rhizophora* and *Bruguiera* mangroves, with the distribution of mangrove species depending on the substrate type profile from sea to land having a different stratification of mangrove species. Salinity distribution and coastal morphology, such as lagoons, shorelines and estuaries, will influence the mangrove species distributed in each area. The abundance of *Nypa fruticans* di Segara Anakan is very large at 4938.25 ha, where the presence of this species is directly adjacent to the river and grass vegetation on the edge of the river, so the abundance and density of *Nypa fruticans* will affect the eel habitat in the area, which is the main route of raya, in addition to disturbance from the Apong net fishing gear (Arai

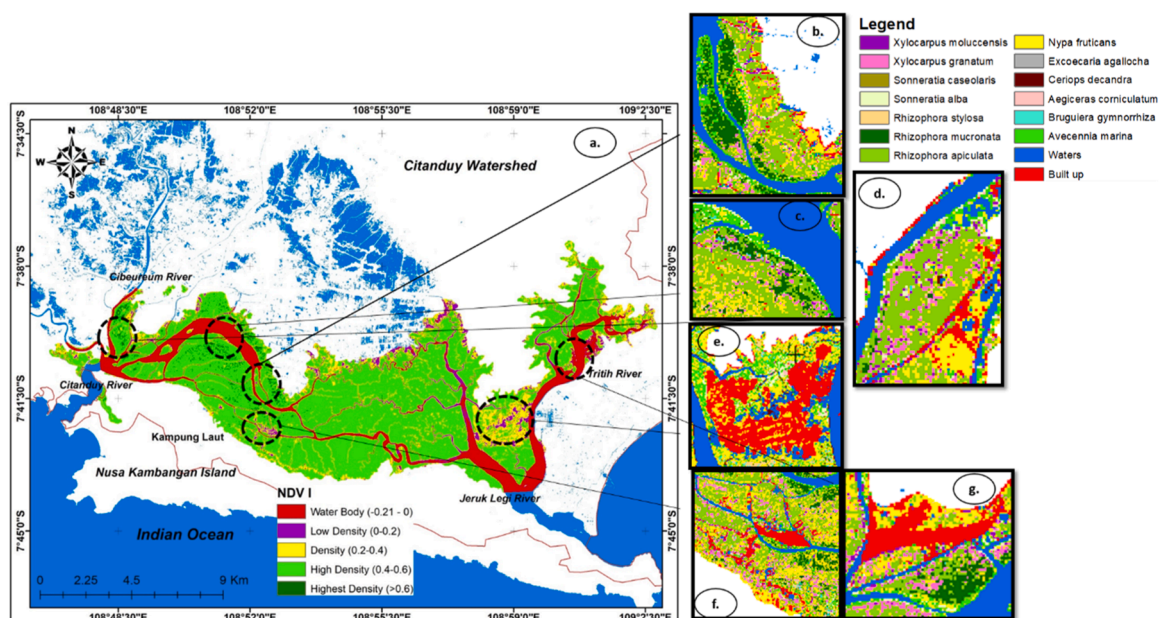


Fig. 7. (a) NDVI map Segara Anakan, (b,c,d) highest density with mangrove species, (e,f,g) low density with mangrove species.

and Chino, 2022). Based on the overlay analysis of NDVI and species distribution, it is known that most mangroves are in the high-density class with a value range of 0.4–0.6, indicating that mangroves in this area are in good condition (Fig. 7c and Fig. 7g) despite experiencing many challenges such as land use change and high sedimentation. There are species well adapted to high sedimentation, namely *Rhizophora mucronata*, and *Rhizophora apiculata*, while on the Citanduy estuary side, downstream areas, especially sedimentation islands, *Xylocarpus granatum* and *Xylocarpus moluccensis*, about 100 m from the riverbank, are still flooded at low tide (Fig. 7d and Fig. 7f) (Basyuni et al., 2023; Chakravorty et al., 2018; Hu et al., 2020).

3.3. Accuracy assessment

The results of the 15-class accuracy classification consisted of 13 types of mangrove species and 2 other land uses using confusion matrix statistical analysis with 169 sample polygons. The main objective of measuring the accuracy of the confusion matrix was to produce a classification with the highest level of accuracy. Based on the accuracy test with a 15 × 15 matrix, the overall accuracy value is 91.28 % with a kappa statistic of 90.15 % (Table 1). UA shows how high and low the accuracy value of the classified species is, based on the results of the analysis of the highest value contained in the species *Xylocarpus moluccensis*, *Xylocarpus granatum*, *Sonneratia caseolaris*, *Sonneratia alba*, *Rhizophora apiculata*, *Bruguiera gymnorhiza*, *Aegiceras corniculatum* with a UA of 100 %. The lowest classification accuracy was found for *Excoecaria agallocha* and *Nypa fruticans*, with UA 66.67 % and 79.31 %, respectively. Classification accuracy for *Excoecaria agallocha* and *Nypa fruticans* species is due to leaf type and density. However, in the field, it was observed that these species' substrate is almost always submerged in water, which will affect NDVI and classification factors due to the reflectance of the submerged substrate. The classification method using SVM can improve the accuracy of the resulting vegetation classification to a large extent compared to supervised/unsupervised classification using either the same training sample or no training sample (Ilham and Marzuki, 2018; Pham and Brabyn, 2017).

The results of the accuracy evaluation (Table 2) showed high classification accuracy of mangrove species with a 15 × 15 classification matrix measuring FoM, O, C and A. The following mangrove species were classified with the highest FoM (over 80 %). *Xylocarpus moluccensis*, *Xylocarpus granatum*, *Sonneratia caseolaris*, *Sonneratia alba*, *Rhizophora stylosa*, *Rhizophora mucronata*, *Ceriops decandra*, *Avecennia marina* and the lowest FoM over 67 %, while *Rhizophora apiculata* had the lowest. Omission also shows a perpendicular relationship with FoM. The lower the omission value, the higher the FoM, indicating that the species has high accuracy. Mangrove species have the highest commission error value, *Rhizophora apiculata* dan *Excoecaria agallocha*, with a value of over 30 % error caused by the low value of FoM, thus increasing the high commission error. Based on the classification results, vegetation index and composite band analysis can detect the presence of mangroves and non-mangroves in detail, as well as the chlorophyll-a content of plants (Kamal et al., 2021).

4. Conclusion

The application of machine learning using high resolution Sentinel 2 satellite imagery can improve the accuracy of classification in the mapping of mangrove species in Segara Anakan, a natural mangrove forest. Using composite bands 8a, 11, 4, NDVI and SAVI can be specific based on colour differences, making it easier to separate mangrove and non-mangrove vegetation pixels. In the upper reaches of the river, where there is anthropogenic land use, mainly coastal village settlements, agriculture and livestock, the dominant mangrove species are *Nypa fruticans*, *Avecennia marina* and *Aegiceras corniculatum*. The lower reaches of the Citanduy estuary and the Jeruk Legi estuary are dominated by *Avecennia marina*, *Bruguiera gymnorhiza* at the estuary edge

and *Nypa fruticans* in the centre of the island. The low and moderate vegetation cover is due to the large amount of land clearing in this area, both for ponds and settlements. Therefore, community empowerment is essential to preserve and maintain the mangrove ecosystem in this area, as it is a source of livelihood for coastal communities.

Author statement

I here by declare that this manuscript is original and has not submitted to other journal for publication

CRediT authorship contribution statement

Anindya Wirasatriya: Writing – review & editing, Methodology, Formal analysis. **Oktavianto Eko Jati:** Writing – original draft, Validation, Data curation. **Arif Rahman:** Validation, Data curation. **Sigit Febrianto:** Writing – original draft, Visualization, Software, Methodology, Data curation, Conceptualization. **Nurul Latifah:** Writing – original draft, Validation, Data curation. **Max Rudolf Muskananfol:** Writing – review & editing, Methodology, Formal analysis, Data curation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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